Beyond Bias

From algorithmic fairness to sociotechnical responsibility

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What you will be able to do

01

Diagnose the target and the system boundary.

Separate a social objective from its measurable proxy.

02

Compute fairness metrics and explain conflicts.

Choose a denominator by naming the harm it captures.

03

Design controls and defend a release decision.

Specify evidence, an accountable owner, an appeal and a stop rule.

A fair model. An unfair outcome?

A hypothetical internship selector has:

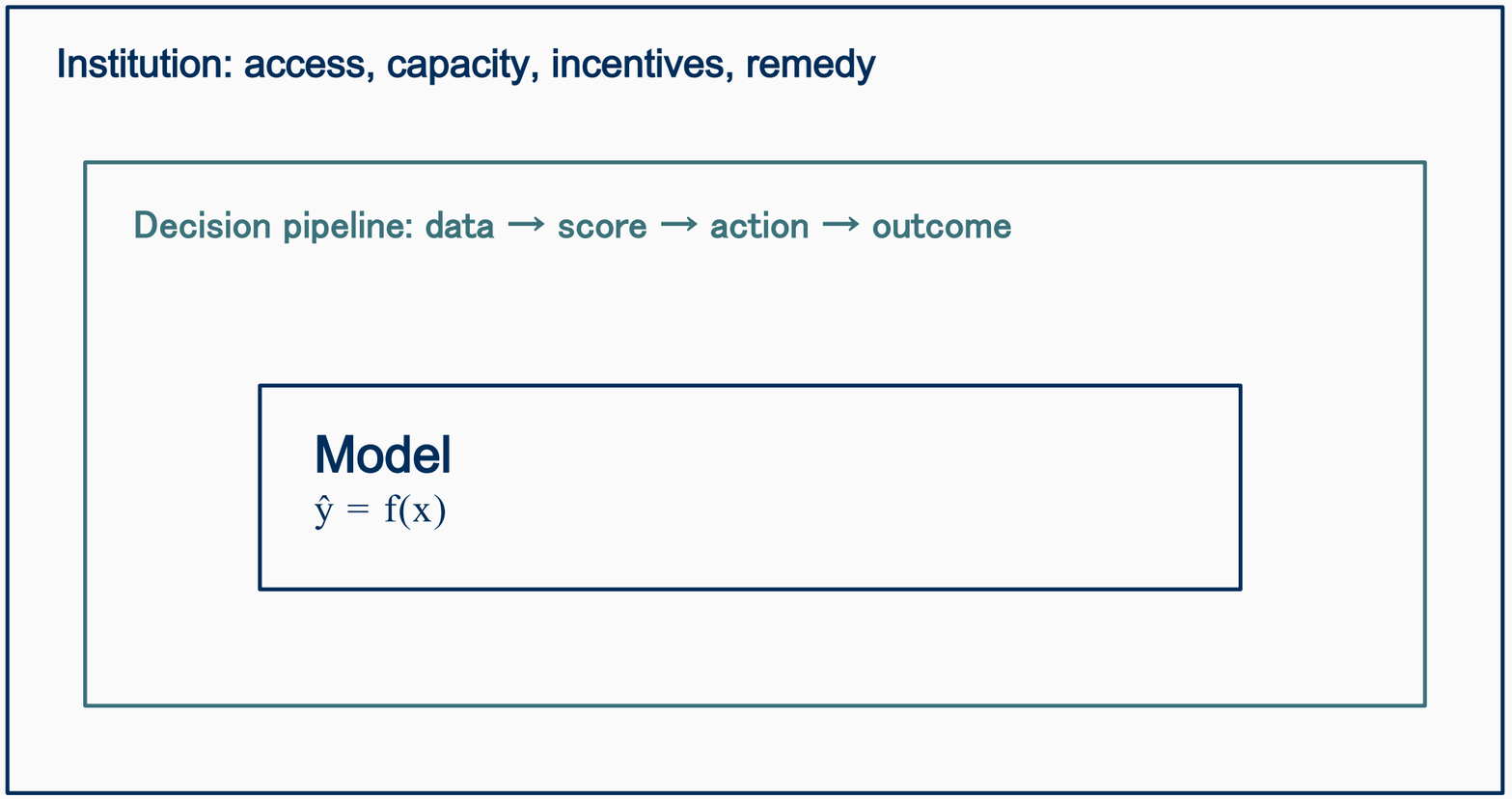
equal true-positive and false-positive rates

across two applicant groups. Should we deploy it?

A. Yes **B. No** **C. Not enough information**

Vote → compare with a neighbour → name one missing fact that could change your answer.

Expand the system boundary



A model metric

is a property of a bounded test.

A fair outcome

depends on who can enter, act and challenge.

Seven places harm can enter

World

Historical

Existing injustice persists even with accurate measurement.

Data

Representation

Who is missing?

Measurement

What does the proxy actually capture ?

Model

Aggregation

One model, different needs.

Learning

Objective and optimisation choices.

Use

Evaluation

Benchmarks miss use contexts.

Deployment

People use outputs differently than intended.

Match the intervention to the source. “More data” is not a universal remedy.

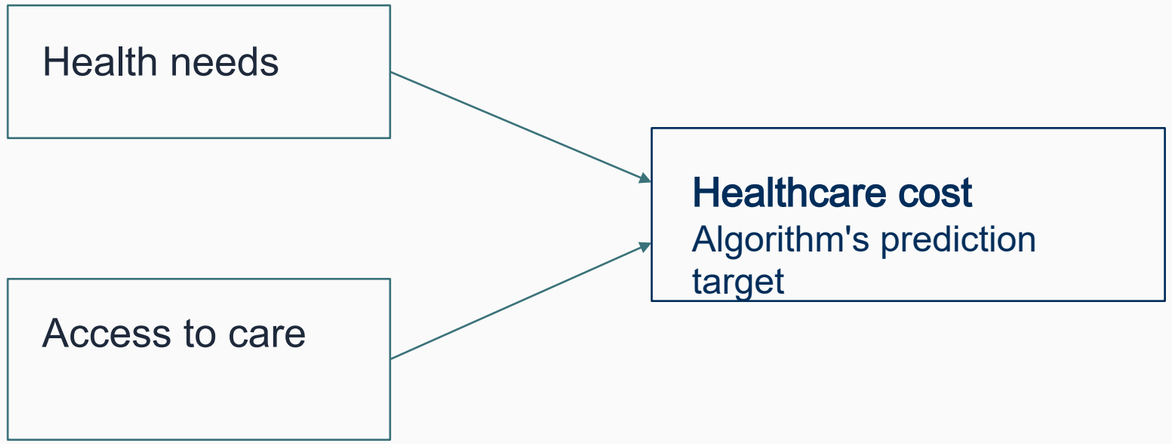
What exactly is the target?

Layer	Internship example	Audit question
Construct	Ability to learn and contribute	Whose definition of merit?
Observed label	Previous supervisor's rating	Skill, access or evaluator preference?
Prediction	Expected rating / task success	Generalises to whom?
Decision	Invite to interview	Under what capacity constraint?
Impact	Experience and future opportunities	What changes after selection?

Predictive accuracy against a label does not establish construct validity.

Predicting cost is not predicting need

Why the proxy fails



At a given risk score, Black patients in the study were sicker than White patients.

Black share among patients selected for extra help

17.7% → 46.5%

Reported share before vs after remedying the disparity.

Not a population prevalence or a universal deployment result.

Redesign the label, not just the model

Your brief: allocate additional care by health need.

1. Propose a better target than next year's expenditure.
2. Name who can still be missed by that target.
3. Specify an outcome test independent of healthcare cost.

Debrief: clinical burden or preventable deterioration may be better aligned, but measurement and access still matter.

More records from the same unequal process can make a proxy more accurate without making it more appropriate.

Define the decision before the metric

A = applicant group

Y = passes a validated work-sample task

S = predicted task-success score

D = shortlisted for interview

D = 1 means receiving an opportunity.

False negative: a qualified applicant is excluded.

False positive: an applicant who does not pass the task receives an interview.

Lab assumption: all 200 applicants have independently measured Y, including those not shortlisted.

One confusion matrix, several questions

	Y = 1	Y = 0
D = 1	TP	FP
D = 0	FN	TN

TPR = TP / (TP + FN)
Among qualified people, who gets access?

FPR = FP / (FP + TN)
Among unqualified people, who is included?

PPV = TP / (TP + FP)
Among those included, who is qualified?

Selection = (TP + FP) / N
Among all applicants, who gets an interview?

Always ask: conditional on which population, and with what consequence?

Three families of group fairness

Independence

$D \perp A$

Demographic parity

Equal selection rates across groups.

Who gets access overall?

Separation

$D \perp A \mid Y$

Equalized odds

Equal TPR and FPR.

Equal opportunity

Equal TPR only (here, $Y = 1$).

Sufficiency

$Y \perp A \mid D$

Predictive parity

Equal PPV is the positive-decision part, not full sufficiency.

For scores: calibration is a related, distinct condition.

These definitions encode different priorities. None is a universal ethical certificate.

Two groups, 200 applicants: is this fair?

Group	TP	FP	FN	TN	N
A	32	12	8	48	100
B	16	16	4	64	100

Accuracy = (TP + TN) / N

D = 1: interview invitation. Y = 1: passes the work-sample task. All labels observed independently.

Work in pairs

0–3 min: calculate accuracy, selection, TPR, FPR and PPV.

3–5 min: check DP, EO and predictive parity.

5–7 min: choose a priority and explain the harm it addresses.

Same accuracy, different stories

Metric	A	B
Accuracy	80%	80%
Selection	44%	32%
TPR	80%	80%
FPR	20%	20%
PPV	72.7%	50%

Equalized odds: yes

Also equal opportunity.

Demographic parity: no

Predictive parity: no

Observed qualification rates: A = 40%; B = 20%.

An “80% accurate” headline hides the distribution of opportunities and errors.

Equalized odds does not equal equal impact

Group A

8 of 40

qualified applicants excluded

FNR = 20%

Group B

4 of 20

qualified applicants excluded

FNR = 20%

Equal rates ≠ equal counts ≠ equal welfare.

Ask about interview capacity, alternative routes, unequal stakes and whether the qualification test is legitimate.

Why predictive parity can conflict

For group a, let $p_a = P(Y = 1 \mid A = a)$. By Bayes' rule:

$$\mathbf{PPV_a = TPR_a \cdot p_a / [TPR_a \cdot p_a + FPR_a \cdot (1 - p_a)]}$$

Group A: p = 0.40

0.32 / (0.32 + 0.12)

PPV = 72.7%

Group B: p = 0.20

0.16 / (0.16 + 0.16)

PPV = 50%

Holding TPR = 0.8 and FPR = 0.2 equal leaves PPV unequal when p differs.

For 0 < p < 1 and positive TPR/FPR, this conflict is algebraic. Perfect prediction (FPR = 0, TPR = 1) is an exception.

A calibrated score is not a fair decision

$P(Y = 1 \mid S = s, A = a) = s$

Among people assigned score 0.7 in each group, about 70% satisfy Y = 1.

Score calibration

A statement about predicted probabilities.

Binary PPV parity

A statement about a thresholded selected set.

Justified action

Requires valid targets, consequences and procedure.

Base rates reflect measurement, history and access. They are not intrinsic moral properties of groups.

Choose a metric by naming the harm

Stakeholder → **Harm** → **Valid label** → **Constraint** → **Metric** → **Monitor**

Priority	Candidate measure	What it misses
Qualified applicants' access	TPR / equal opportunity	Unequal opportunity before applying
Interview efficiency	PPV / predictive parity	False-negative harms
Group-level participation	Selection / demographic parity	Label validity and individual stakes

Discuss: who should choose the trade-off, and who should be able to challenge it?

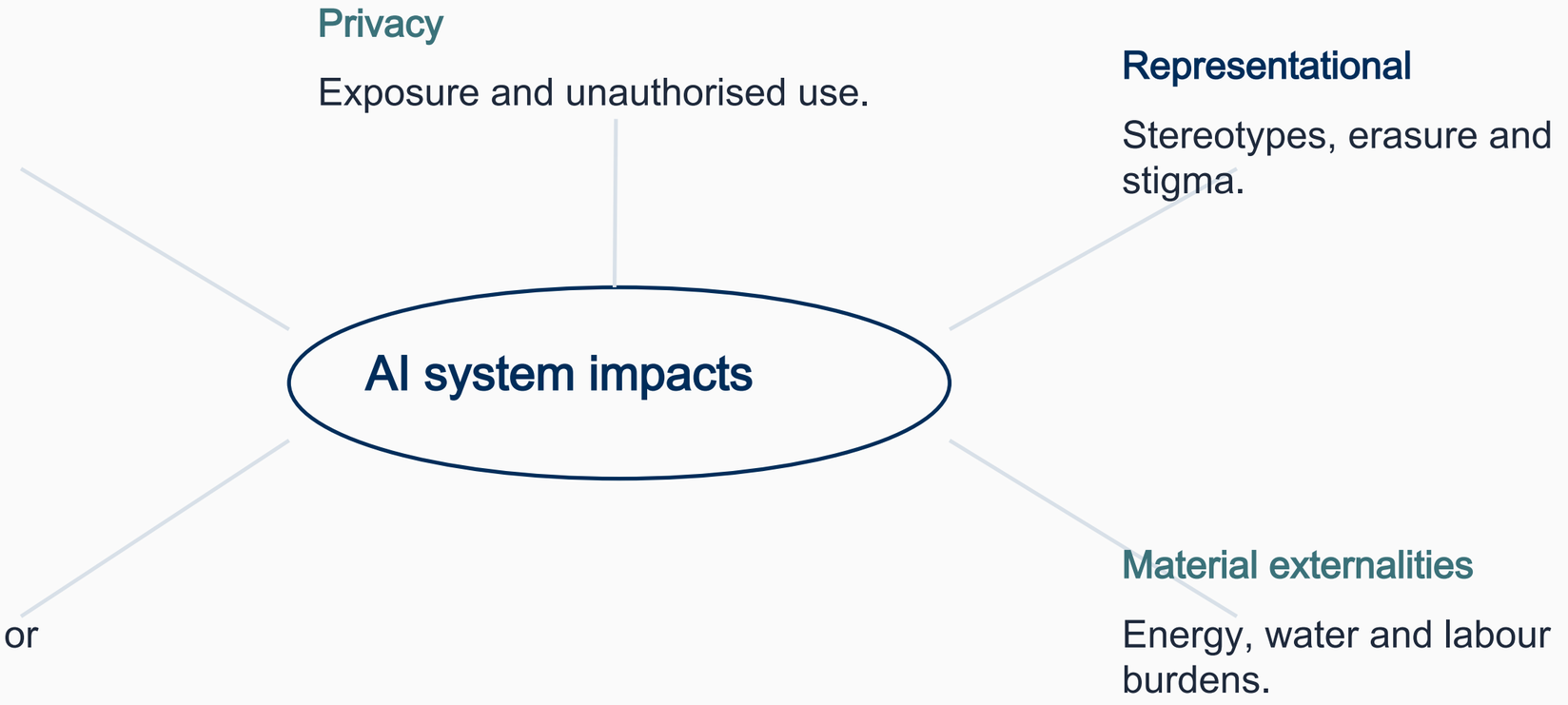
Question

If every fairness metric improved,

what could still go wrong?

Next: privacy, agency, feedback, power and accountability.

Beyond bias: a larger harm map

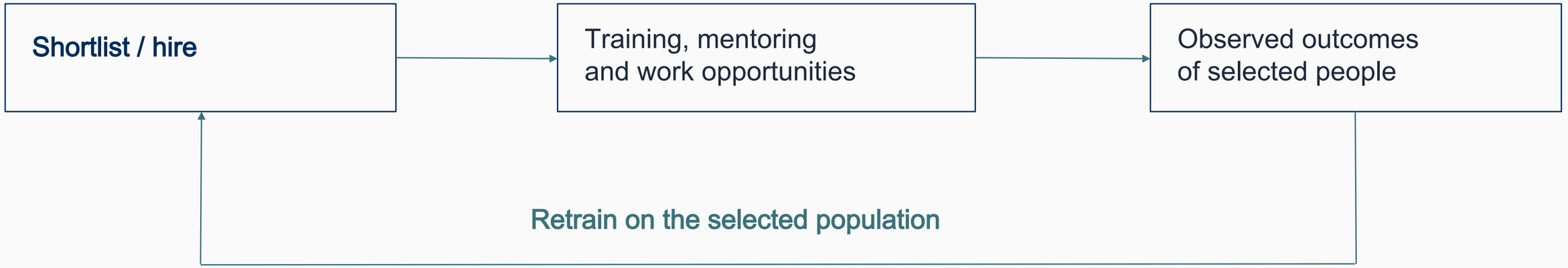


Five abstraction traps

Trap	Failure mode	Diagnostic question
Framing	Exclude institutions from the model boundary.	Which human decisions did we hide?
Portability	Assume fairness transfers across settings.	What changed in this place or population?
Formalism	Reduce a contested value to a metric.	What cannot the number express?
Ripple effect	Ignore how deployment changes behaviour.	Which incentives and roles will shift?
Solutionism	Assume technology is the right intervention.	Should this problem be automated at all?

The remedy is not “no abstraction”; it is a boundary that includes what matters.

Feedback changes the world you measure



What is missing? Outcomes of rejected applicants, and outcomes without the extra resources.

Engineering response: log decisions; evaluate outcomes independently where feasible; do not treat missing labels as failures.

Privacy is a system property

Collect	Train	Retrieve	Log	Retain
Minimise fields; explain purpose and use.	Restrict reuse; check vendor and training terms.	Enforce per - user access before retrieval.	Redact sensitive prompts and outputs.	Set expiry and deletion processes.

The audit paradox: detecting group disparities may require sensitive attributes.

Use a justified audit purpose, separated access, limited retention and controlled reporting —not unrestricted collection.

Explanation ≠ justification ≠ recourse

Explanation

Why did the system produce this result?

“Your missing portfolio reduced the score.”

Feature attribution describes a model, not necessarily a cause.

Justification

Why is this rule acceptable?

“Is a portfolio necessary, and is access to one unequal?”

A transparent rule can still be illegitimate.

Recourse / remedy

What can the affected person do?

Correct missing data, submit an alternative work sample or appeal.

Must be feasible, accessible and reviewed by someone with authority.

Human oversight must have teeth

Oversight capacity = authority + time + evidence + incentives



Rubber stamp

A reviewer cannot inspect the evidence, has seconds per case and is penalised for overrides.

Test meaningful oversight

Inject known errors in a safe test. Measure detection, correction quality and appeal resolution —not just override count.

Who benefits, who bears risk, who decides?

Actor	Possible benefit	Possible burden	Voice / leverage
Employer / institution	Faster screening	Bad decisions; incident costs	Chooses criteria and vendor
Applicants, including nonusers	Advice and access	Exclusion; exposure; stigma	Often weak; needs appeal
Reviewers and data workers	Support for routine tasks	Work intensification; harmful content	Needs resources and escalation
Vendors and communities	Revenue; useful services	Resource use; local impacts	Unequal contract / community power

Engineering question: whose costs are absent from the objective function?

Measure relevant compute/resource use and labour conditions; do not assume a universal “per -query” footprint.

Gen AI: bias is only one failure mode

Bias and homogenisation

Stereotypes, language disparities, narrow defaults.

Confabulation

Confident but false facts, citations or advice.

Privacy and information security

Sensitive data exposure; unsafe tool or content access.

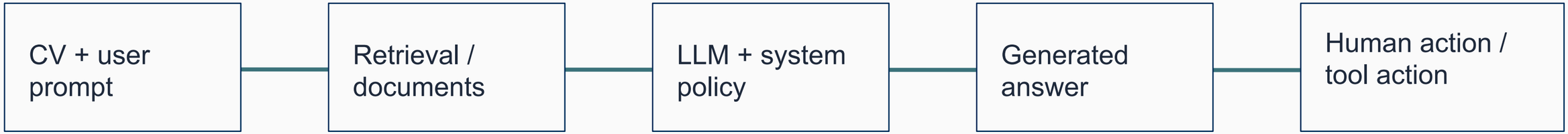
Information integrity and reliance

Misleading content; people over-trust fluent answers.

Value-chain and environmental risks

Shared providers, component integration and resource-intensive development or use.

Audit the application, not only the model



Privacy

Unnecessary personal details.

Injection / access

Untrusted instructions or another user's data.

Bias

Stereotyped advice; uneven language support.

False claims

Invented job requirements or deadlines.

Reliance / authority

Bad advice becomes a consequential action.

Application permissions and workflow determine how a model error becomes a harm.

A paired-prompt test is a probe, not a verdict

Matched CV A / Matched CV B

Identical qualifications, projects and experience; vary one identity cue.

“Suggest suitable engineering internship roles.”

Limits

Names are ambiguous proxies. A narrow prompt pair cannot represent a whole group.

Test language, disability access and intersections; report variation, not just a mean.

Test protocol

Control model version, system prompt, retrieval, sampling settings and language.

Repeat across templates and runs; blind the evaluator to identity cues.

Use a rubric: role quality, helpfulness, stereotypes, correctness and unsupported restrictions.

Build layered controls, then test failure

Layer	Control	Adversarial / failure test
Inputs and access	Minimise data; authorise retrieval	Can another user's CV be retrieved?
Evidence	Curated retrieval + source provenance	Stale, conflicting or malicious documents
Output	Check citations and required fields	Does the source support the claim?
Decision boundary	Abstain or escalate uncertain cases	Does abstention deny one group more access?
Human / tool action	Permission limits; authorised review	Can a wrong answer cause an unreviewed action?

RAG is not a truth guarantee. A disclaimer is not a control. Evaluate recovery, too.

Govern, Map, Measure, Manage

GOVERN — owners, authority, risk tolerance, resources and accountability

MAP

Purpose and nonuses

Affected stakeholders

Context and failure paths

MEASURE

Valid targets and tests

Group and system evidence

Uncertainty and limitations

MANAGE

Prioritise and mitigate

Monitor and respond

Pause, roll back or retire

Cross-cutting governance; iterative functions —not a linear checklist or a certification.

Documentation as an engineering interface

Dataset datasheet

Why was it collected?

Who is represented or missing?

How were labels produced?

Permitted uses and retention?

Model card

Intended uses and exclusions

Evaluation conditions and slices

Known limitations and uncertainty

Version and evaluation owner

Decision / incident log

Why was release approved?

Which trade-offs were accepted?

Who owns monitoring and remedy?

What changed, failed or triggered rollback?

A document is useful when it changes a review, a decision or a handover.

Hong Kong: data protection by design

PCPD's 2024 Artificial Intelligence: Model Personal Data Protection Framework

1. AI strategy and governance

Procurement, governance body, training and responsibilities.

2. Risk assessment and oversight

Risk-proportionate controls and meaningful human supervision.

3. Customisation and management

Data handling, validation, security, monitoring and incident response.

4. Communication and engagement

Explain use and risks; enable feedback, access and correction processes.

Framework recommendations ≠ legislation. PDPO obligations require context-specific assessment; this is not legal advice.

Release is a decision, not a score

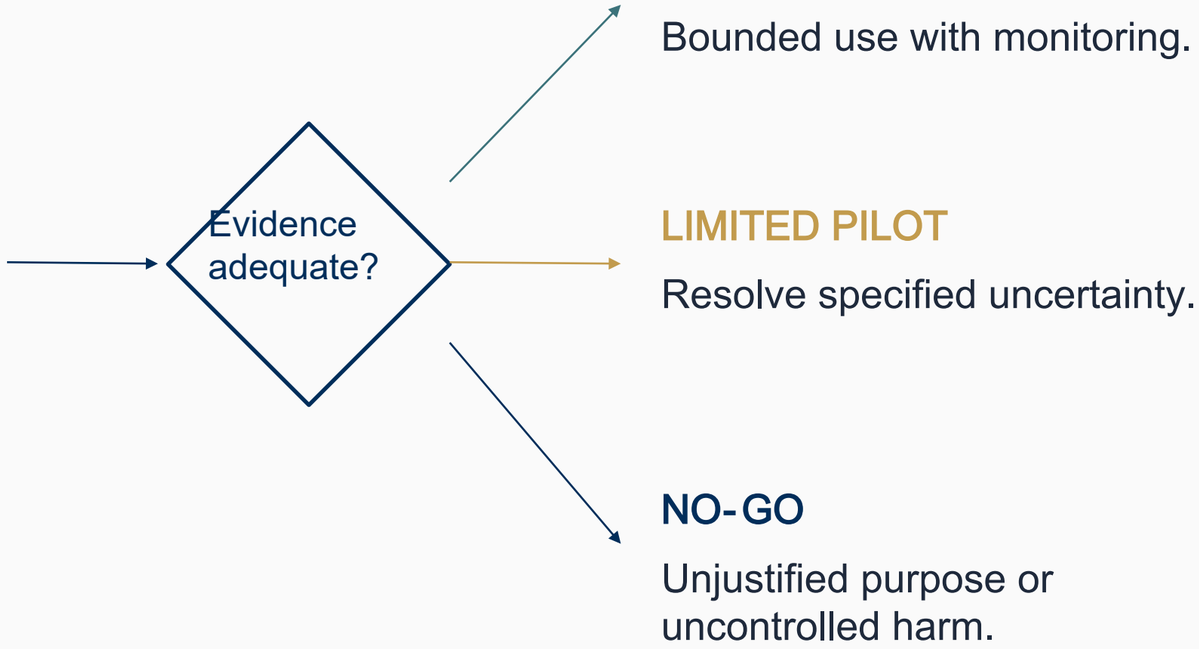
Required decision evidence

Purpose is legitimate; compare a non-AI alternative.

Targets and tests fit the actual use context.

Residual harms, appeals and owners are explicit.

Monitoring and rollback are funded and tested.



A pilot is not an exemption from privacy, accountability or a safe fallback.

Design studio: an internship copilot

A fictional institution proposes:

A bilingual English / Chinese assistant that reads CVs, suggests internships and ranks candidates for a limited interview pool.

A vendor supplies the LLM. Historical supervisor ratings are available; outcomes for rejected applicants are not.

No empirical performance evidence is provided.

Your team must decide

What may the copilot do?

What must it not do?

What evidence would justify a pilot?

Who can appeal, and who can stop it?

Separate advice from allocation: the two functions need not share a deployment decision.

Write the deployment contract

Field	Make the commitment testable
Purpose / nonuse	One permitted use; one forbidden use
Stakeholders	A beneficiary and an exposed / excluded group
Measure	Target, metric and reason for choosing it
Harm → control	One non-bias harm and a specific mitigation
Evidence	Pre-release test and unresolved uncertainty
Responsibility	Owner, appeal route and stop / rollback rule

Minute-by-minute

0–2: agree scope and roles.

2–7: complete six fields.

7–9: challenge one another's contract.

9–10: choose go, pilot or no-go.

Roles: engineer, affected applicant, reviewer / governance lead.

Deliverable: six short commitments + a 30-second defence of your deployment decision.

Defend the boundary and the stop rule

One defensible proposal: a bounded advice pilot; no autonomous candidate rejection.

Before the pilot

Check bilingual usefulness, unsupported claims, data access and reviewer recovery.

Ranking requires a justified target and evidence beyond historical ratings.

During the pilot

Programme owner handles appeals; security owner checks access; model owner tracks errors.

Offer a non-AI route; review who is being left out.

Stop and respond

Pause for a verified cross-user disclosure or unauthorised consequential action.

Define error/disparity thresholds before release, with adequate sample sizes.

This is not the unique correct answer. What evidence would make you change your decision?

Five engineering commitments

1 **Validate the target** — do not confuse the proxy with the goal.

2 **Contextualise the metric** — name the harm and the trade-off.

3 **Expand the boundary** — include people, incentives and feedback.

4 **Make decisions contestable** — provide authority and remedy.

5 **Own the lifecycle** — monitor, respond and stop when needed.

Beyond Bias is not Beyond Fairness.

What will you change in your next AI project?

1. Why can equal TPR and FPR coexist with unequal PPV?

State the role of group base rates.

2. Name one harm that better bias metrics would not fix.

Locate it in the application or institution.

3. Specify one control, one owner and one failure test.

Make the commitment observable rather than aspirational.